



ASUS touchpad has a screen

ASUS is releasing a new laptop where the touchpad will double as a 5.5inch screen known as the ScreenPad. <https://digit.in/jul18-29-1>



Cortana can hack PCs

McAfee security researchers confirm that hackers can use Cortana from lock screen to execute malicious software. <https://digit.in/jul18-29-2>

NETGEAR Nighthawk Pro XR500 Gaming Router

A new entrant in the gaming peripheral scene

The Netgear XR500 is being marketed as a “gaming router” given the built-in QoS settings that it sports thanks to their partnership with DumaOS. Here’s how the Netgear XR500 stacks up against the competition.

Packaging

The Nighthawk series from Netgear have always sported sharp edges and the XR500 continues that style philosophy. The indicator lights have been moved forward compared to the R7000 but the WPS and Wi-Fi on/off switches are still on the top. We’d have preferred that these buttons be placed on the side to avoid jamming due to dust accumulation.

The XR500 sports four external antennae which are all removable, however, the connectors on the sides are slightly recessed. The antennae connectors are quite rigid.

The rear of the router houses 4 Gigabit LAN and 1 Gigabit WAN ports, along with the power socket. The router does have ventilation slits on the front and the sides of the rear face.

At the heart of the XR500 lies the IPQ8065 from Qualcomm which has been built for home networking products. It combines a dual core Qualcomm Krait CPU clocked at 1.7 GHz along with a dual-core 800 MHz NSS, with the best interface support.

For the radios, the XR500 uses two QCA9984 SoCs which are Wave-2 802.11ac radios that supports a dual band 4T4R setup with peak data rates of up to 1.7 Gbps and also has support for MIMO. At close range, you can switch to a higher frequency channel to increase throughput considerably. For the 2.4 GHz radio, we see the Sky-



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works 2623L being used for each of the four streams and on the 5 GHz radio, Netgear has opted for the rfmd 5542 power amplifiers. In either case, we see that the amps are optimised for space constraints and offer some of the best performance characteristics.

All sensitive SoCs are well shielded and the components are spaced well-apart to help with reducing hotspots. The single 470uF electrolytic filter cap at the beginning should have been replaced with a solid state capacitor for better life.

Features

There’s the Geo-Filter feature which lets you reduce routing distance by geofencing your router. While a hard and fast ping limit isn’t there, you can prevent low ping players from discovering your servers. Either you set generic rules or have IP specific blacklists to avoid known culprits with heavy lag. Moreover, developers who’ve publicly released their server IPs have allowed for listings in the UI.

Other features which stand out are the anti-bufferbloat and built-in deep-packet inspection features. The Netgear XR500 certainly takes the firmware game up a notch and in the right direction – mostly for enthusiasts though. On the hardware front, the

Netgear XR500 sports two USB 3.0 ports which are now common in this particular price bracket.

Performance

For the two USB 3.0 ports, in our read tests, we could manage about 108 MB/s while for write operations we could hit roughly half that speed at 53 MB/s. In our wired throughput test for the LAN-LAN configuration, the XR500 is well capable of pushing about 90 MB/s for read operations and roughly 66 MB/s for the write cycle.

Moving on to the individual wireless bands, we ran a 3x3 throughput test with the router. On the 5 GHz band, we managed about 440 Mbps on the uplink and around 370 Mbps on the downlink. When the same test was run on the 2.4 GHz band, we then moved onto the individual bands. On the 2.4 GHz band, we managed to get an average speed of 80 Mbps on the downlink and around 85 Mbps on the uplink. In our video streaming tests, the router was well capable of playing 3x high bit-rate video streams averaging about 30 Mbits each.

In our gaming tests, we compared the router against some of the popular game VPN services such as WTFast and found them to be more or less the same. Essentially, you’re getting performance equivalent of a paid software with the Netgear XR500 and it’s “gaming” router moniker holds true.

Verdict

For the price of ₹23,000 the Netgear XR500 isn’t exorbitantly priced. It should be noted that the router’s features are better left to enthusiasts to play with because the average gamer is simply not educated about the advanced networking features that the XR500 ships with.

—Mithun Mohandas

SPECIFICATIONS

SOC: Qualcomm IPQ8065 | WLAN1: Qualcomm QCA9984 | WLAN2: Qualcomm QCA9984 | LAN: 4x Gigabit Ethernet | WAN: 1x Gigabit Ethernet | ANTENNAE: 4x External | WPS: Yes | USB PORTS: 2 | PROTOCOLS: 802.11ac (1900 Mbps)

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